Market Equilibrium: Stability and Time Paths

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Introduction

The equilibrium price of a market is the price at which the quantity buyers want to purchase equals the quantity sellers want to supply. Finding that price is a static question. It does not say whether the market will ever reach that price, or what happens after a shock such as a bad harvest or a sudden change in demand moves the price away from it. Those are dynamic questions, and they matter more in practice, because an equilibrium that the market drifts away from tells us very little about the prices we will actually observe.

To answer them, we model price as a function of time, $p(t)$, and describe how it changes with a differential equation. The equilibrium price p^* is a constant solution of that equation. We call p^* *stable* if paths that start near it move toward it over time, and *unstable* if they move away.

This paper compares two models built on the same linear demand and supply curves. In the first, the price adjusts in proportion to excess demand: when buyers want more than sellers offer, the price rises, and when sellers offer more than buyers want, it falls. In the second, the market clears at every instant, but buyers and sellers react to the direction the price is moving as well as its level. Parts (a) and (b) solve the first model and prove that its equilibrium is globally stable. Parts (c) and (d) solve the second model and show that its equilibrium is unstable. Part (e) applies the second model to a numerical example and interprets the result. The main finding is that the two models share the same equilibrium price but have opposite stability, so an equilibrium price alone does not tell us how a market behaves over time.

Model and Equilibrium

Let $p_0 = p(0)$. All model parameters are positive, and $t \geq 0$. For the linear demand and supply functions

$$q_d = d_0 - d_1 p, \qquad q_s = -s_0 + s_1 p,$$

equating demand and supply gives

$$d_0 - d_1 p^* = -s_0 + s_1 p^* \quad \Longrightarrow \quad p^* = \frac{d_0 + s_0}{d_1 + s_1}.$$

Figure 1 shows this equilibrium for one choice of parameters. At any price below p^*, demand exceeds supply; at any price above p^*, supply exceeds demand. The difference $q_d - q_s$ is the *excess demand*, and it drives the adjustment model in part (a).

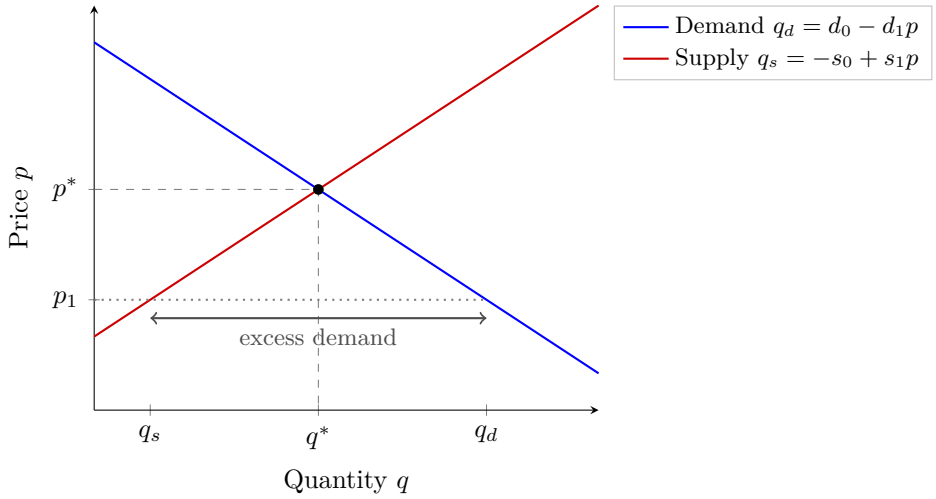


Figure 1: Linear demand and supply (illustrated with $d_0 = 10$, $d_1 = 1$, $s_0 = 2$, $s_1 = 1$, so $p^* = 6$). At the price $p_1 < p^*$, buyers demand more than sellers supply.

(a) Price adjustment path

Substitute the demand and supply functions into the adjustment equation:

$$\begin{aligned} p'(t) &= \lambda(q_d - q_s) \\ &= \lambda\big[d_0 + s_0 - (d_1 + s_1)p(t)\big] \\ &= -\lambda(d_1 + s_1)\big[p(t) - p^*\big]. \end{aligned}$$

Set $x(t) = p(t) - p^*$ and $c = -\lambda(d_1 + s_1)$. Since p^* is constant, the equation becomes $x' = cx$. Multiplying by the integrating factor e^{-ct} gives

$$\frac{d}{dt}\big(e^{-ct}x(t)\big) = 0 \quad \Longrightarrow \quad x(t) = x(0)e^{ct} = (p_0 - p^*)e^{ct}.$$

Therefore the required price path is

$$p(t) = \big[p(0) - p^*\big]e^{ct} + p^*, \qquad c = -\lambda(d_1 + s_1). \tag{1}$$

This also includes the constant solution $p(t) \equiv p^*$ when $p_0 = p^*$.

(b) Global stability

Because $\lambda, d_1, s_1 > 0$, we have $c < 0$, so $e^{ct} \to 0$. Consequently,

$$\lim_{t\to\infty} p(t) = p^* + (p_0 - p^*) \lim_{t\to\infty} e^{ct} = p^* \qquad \text{for every initial price } p_0.$$

Therefore, the market equilibrium is globally stable. Prices below p^* rise toward it, and prices above p^* fall toward it. The distance from equilibrium decays exponentially, without overshooting. Figure 2 shows several paths with different starting prices.

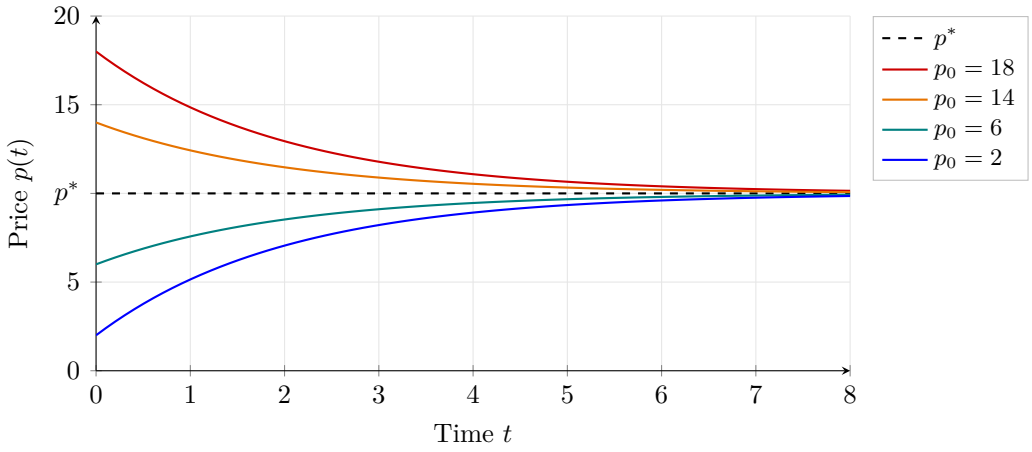


Figure 2: Price adjustment paths $p(t) = p^* + (p_0 - p^*)e^{ct}$ for several initial prices (illustrated with $p^* = 10$ and $c = -0.5$). Every path converges to p^*.

(c) Market-clearing time paths

In the expectations model, demand and supply depend on both the price and its rate of change:

$$q_d(t) = d_0 - d_1 p(t) + d_2 p'(t), \qquad q_s(t) = -s_0 + s_1 p(t) - s_2 p'(t).$$

A market-clearing path satisfies $q_d(t) = q_s(t)$ at every time. A dynamic equilibrium additionally has $p'(t) = 0$, giving the same constant equilibrium price p^*.

Equate demand and supply and collect terms:

$$\begin{aligned} d_0 - d_1 p + d_2 p' &= -s_0 + s_1 p - s_2 p', \\ (d_2 + s_2)p' &= (d_1 + s_1)p - (d_0 + s_0). \end{aligned}$$

Define the positive constant

$$k = \frac{d_1 + s_1}{d_2 + s_2} > 0.$$

Since $d_0 + s_0 = (d_1 + s_1)p^*$, the differential equation is

$$p' = k(p - p^*).$$

Again let $x = p - p^*$. The equation $x' = kx$, with $x(0) = p_0 - p^*$, has solution $x(t) = (p_0 - p^*)e^{kt}$. Thus

$$p(t) = p^* + (p_0 - p^*) \exp\left(\frac{d_1 + s_1}{d_2 + s_2} t\right). \tag{2}$$

(d) Stability of market-clearing paths

Not every market-clearing path converges to equilibrium. The exponential in (2) grows because $k > 0$. The long-run behavior depends on the initial price:

$$\begin{aligned} p_0 > p^* \quad &\Longrightarrow \quad p(t) \to +\infty, \\ p_0 = p^* \quad &\Longrightarrow \quad p(t) = p^* \text{ for all } t \geq 0, \\ p_0 < p^* \quad &\Longrightarrow \quad p(t) \to -\infty. \end{aligned}$$

For any nonzero initial displacement,

$$|p(t) - p^*| = |p_0 - p^*|e^{kt}.$$

Even an arbitrarily small departure therefore grows over time. The equilibrium is unstable, and only a path that starts exactly at p^* converges to p^*. Figure 3 shows this: paths that begin close to p^* stay near it briefly, then move away at an increasing rate.

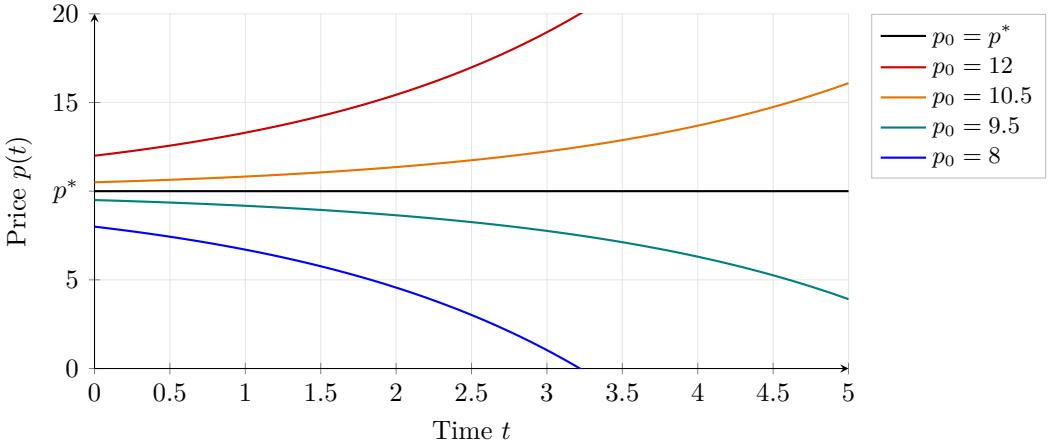


Figure 3: Market-clearing paths $p(t) = p^* + (p_0 - p^*)e^{kt}$ (illustrated with $p^* = 10$ and $k = 0.5$). Only the path starting exactly at p^* stays there; every other path moves away from equilibrium.

The difference in stability follows from the sign of the coefficient of $p - p^*$. The first model gives $p' = -\lambda(d_1 + s_1)(p - p^*)$, which pushes prices toward equilibrium. Market clearing with expectations gives $p' = k(p - p^*)$, which pushes prices away from equilibrium. Equality of supply and demand does not by itself imply stability.

(e) Numerical application

The initial price is $p(0) = \$5$, and t is measured in months. The given functions are

$$q_d(t) = 30 - 2p(t) + 4p'(t), \qquad q_s(t) = -20 + p(t) - 6p'(t).$$

Market clearing requires

$$30 - 2p + 4p' = -20 + p - 6p' \quad \Longrightarrow \quad 10p' = 3p - 50.$$

Hence $p^* = 50/3$, $k = 3/10$, and $p_0 - p^* = 5 - 50/3 = -35/3$. Using (2),

$$p(t) = \frac{50}{3} - \frac{35}{3}e^{3t/10}. \tag{3}$$

Price after 10 months. Substitute $t = 10$:

$$p(10) = \frac{50 - 35e^3}{3} \approx -\$217.66.$$

Long-run price and rate of change. As $t \to \infty$, $e^{3t/10} \to \infty$. Thus, differentiating (3),

$$p(t) \to -\infty, \qquad p'(t) = -\frac{7}{2}e^{3t/10} \to -\infty.$$

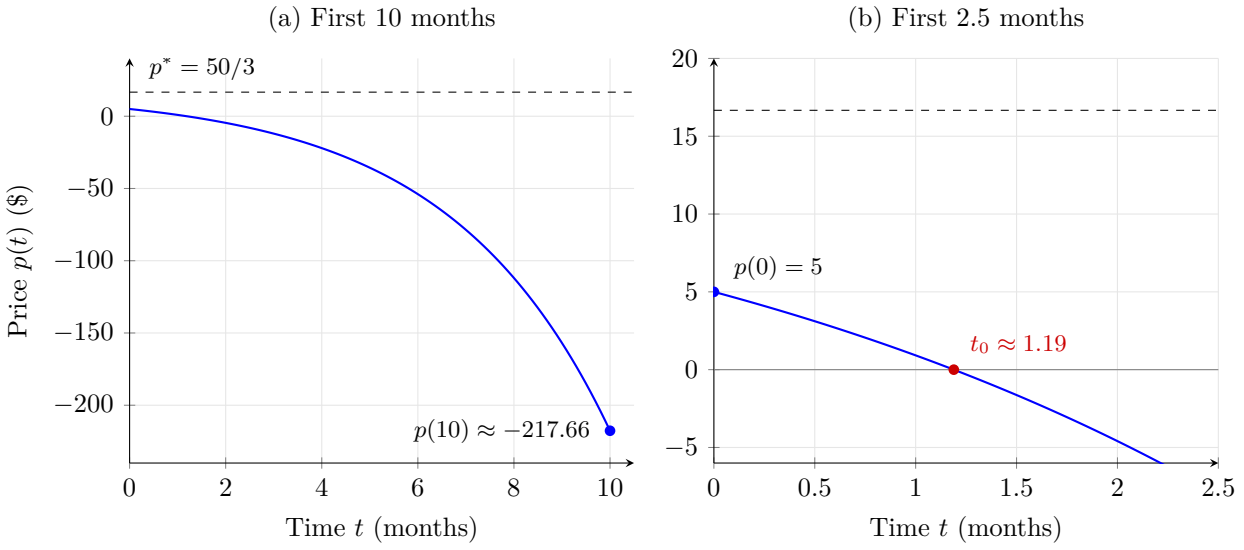


Figure 4: The numerical market-clearing path $p(t) = \frac{50}{3} - \frac{35}{3}e^{3t/10}$. The price starts below p^* and falls away from it, reaching zero at $t_0 = \frac{10}{3}\ln\frac{10}{7} \approx 1.19$ months.

Since $p''(t) = -(21/20)e^{3t/10} < 0$, the price falls at an increasingly rapid rate. The rate changes from $-\$3.50$ per month initially to approximately $-\$70.30$ per month after 10 months. Figure 4 plots the price path over the full 10 months and over the first two and a half months.

Evolution of expectations. Under the model's interpretation, buyers and sellers anticipate further price decreases. Since $d_2 p' = 4p' < 0$, buyers postpone purchases to benefit from lower future prices, reducing current demand relative to its value at the same price without expectations. Since $-s_2 p' = -6p' > 0$, sellers offer more now to avoid receiving an even lower price later. Both effects strengthen as p' becomes more negative.

The total quantity nevertheless increases along the path: $q_d(t) = q_s(t) = -10/3 + (28/3)e^{3t/10}$. The effect of the falling price on demand outweighs buyers' incentive to wait.

The formal solution reaches zero at $t_0 = (10/3)\ln(10/7) \approx 1.19$ months. If prices must remain nonnegative, the model is economically applicable only through t_0. The value $-\$217.66$ is the model's mathematical prediction after 10 months; an ordinary market with nonnegative prices would require a different model beyond the zero-price point.

Conclusion

Both models in this paper use the same linear demand and supply curves, so both have the same equilibrium price $p^* = (d_0 + s_0)/(d_1 + s_1)$. What separates them is how the price behaves away from that point.

In the price adjustment model, excess demand pushes the price up and excess supply pushes it down. The deviation from equilibrium decays at the rate $\lambda(d_1 + s_1)$, so every path converges to p^*, regardless of where it starts. Buyers and sellers who are more sensitive to price (larger $d_1 + s_1$), or a market that adjusts faster (larger λ), make the convergence quicker. In the market-clearing model with expectations, buyers and sellers respond to the direction the price is moving. A falling price leads buyers to wait and sellers to sell now, and both reactions push the price down further. The deviation grows at the rate $k = (d_1 + s_1)/(d_2 + s_2)$, so the equilibrium is unstable: any initial price

other than p^* moves away from it. The stronger the reaction to price changes ($d_2 + s_2$), the smaller k is and the more slowly this happens, but it never stops.

The numerical example shows how fast the instability takes hold. Starting from \$5, below the equilibrium of about \$16.67, the price falls faster every month and reaches zero after about 1.19 months. The prediction of $-\$217.66$ after 10 months is not a realistic price; it shows that the model stops describing a real market once the price hits zero.

The broader lesson is that solving $q_d = q_s$ identifies an equilibrium but says nothing about whether a market will reach it. That depends on the dynamics, which here come down to the sign of a single coefficient. Our models are simplified: the curves are linear, the parameters are constant, and nothing keeps the price from becoming negative. More realistic models could let expectations adapt over time, add a price floor, or combine adjustment toward excess demand with expectation effects. Even in these simple forms, though, the comparison shows that self-reinforcing expectations can make an otherwise ordinary market unstable.